

UNIT I CLOUD ARCHITECTURE MODELS AND INFRASTRUCTURE

Cloud Architecture: System Models for Distributed and Cloud Computing – NIST Cloud Computing Reference Architecture – Cloud deployment models – Cloud service models; Cloud Infrastructure: Architectural Design of Compute and Storage Clouds – Design Challenges

1.1. CLOUD ARCHITECTURE: SYSTEM MODELS FOR DISTRIBUTED AND CLOUD COMPUTING

- Distributed and cloud computing systems are built over a large number of autonomous computer nodes. These node machines are interconnected by SANs, LANs, or WANs in a hierarchical manner.
- With today's networking technology, a few LAN switches can easily connect hundreds of machines as a working cluster.
- A WAN can connect many local clusters to form a very large cluster of clusters. In this sense, one can build a massive system with millions of computers connected to edge networks.
- Massive systems are considered highly scalable, and can reach web-scale connectivity, either physically or logically.
- In Table 1.1, massive systems are classified into four groups: clusters, P2P networks, computing grids, and Internet clouds over huge data centers. In terms of node number, these four system classes may involve hundreds, thousands, or even millions of computers as participating nodes.
- These machines work collectively, cooperatively, or collaboratively at various levels. The table entries characterize these four system classes in various technical and application aspects.

Functionality, Application	Computer Cluster	Peer to Peer Networks	Data/Computational Grids	Cloud Platforms
Architecture, Network Connectivity and Size	Network of computer nodes interconnected by SAN, LAN OR WAN hierarchically	Flexible network of client machines logically connected by an overlay network	Heterogeneous clusters interconnected by high-speed network links over selected resource sites.	Virtualized cluster of server over data centers via SLA.
Control and Resources	Homogeneous nodes with distributed control, running UNIX or LINUX	Autonomous client nodes free in and out, with self-Organization	Centralized control, server-oriented with authenticated security.	Dynamic resource provisioning of server storage and networks.
Application and Network-Centric Services	High-performance computing, search	Most appealing to business file sharing, content delivery and social networking.	Distributed super-computing, global	Upgraded web search, utility

	ch Engine and web services etc.		problem solving and data center services.	computing and sourced computing services.
Representative Operational System	Google search engine,Sun Blade,IBM Road Runner,Cray XT4 etc.	Gnutella,mule,BiTorrent,Napster,Skype, JXTA	Tera grid,GriPhyN,UK EGEE,D-Grid,ChinaGrid etc.	Google App Engine,IBM Bluecloud,A WS and Microsoft Azure

- From the application perspective, clusters are most popular in supercomputing applications. In 2009, 417 of the Top 500 supercomputers were built with cluster architecture.
- It is fair to say that clusters have laid the necessary foundation for building large-scale grids and clouds.
- P2P networks appeal most to business applications. However, the content industry was reluctant to accept P2P technology for lack of copyright protection in ad hoc networks.
- Many national grids built in the past decade were underutilized for lack of reliable middleware or well-coded applications.
- Potential advantages of cloud computing include its low cost and simplicity for both providers and users

1.1.1 Clusters of Cooperative Computers

A computing cluster consists of interconnected stand-alone computers which work cooperatively as a single integrated computing resource. In the past, clustered computer systems have demonstrated impressive results in handling heavy workloads with large data sets.

Cluster Architecture

- Figure 1.1 shows the architecture of a typical server cluster built around a low-latency, high-bandwidth interconnection network.
- This network can be as simple as a SAN (e.g., Myrinet) or a LAN (e.g., Ethernet). To build a larger cluster with more nodes, the interconnection network can be built with multiple levels of Gigabit Ethernet, Myrinet, or InfiniBand switches.
- Through hierarchical construction using a SAN, LAN, or WAN, one can build scalable clusters with an increasing number of nodes.
- The cluster is connected to the Internet via a virtual private network (VPN) gateway. The gateway IP address locates the cluster. The system image of a computer is decided by the way the OS manages the shared cluster resources. Most clusters have loosely coupled node computers.

- All resources of a server node are managed by their own OS. Thus, most clusters have multiple system images as a result of having many autonomous nodes under different OS control.

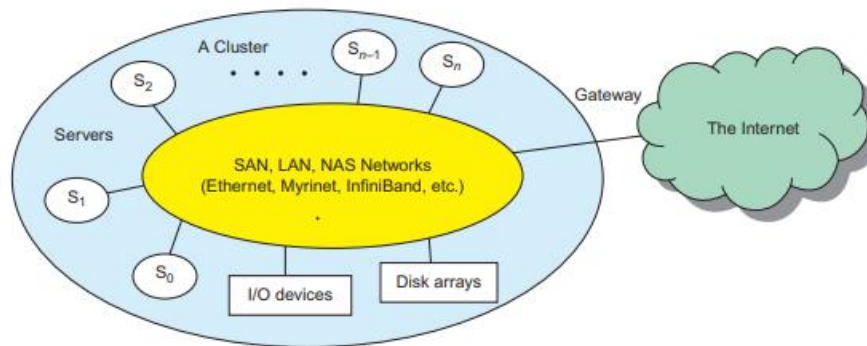


Figure 1.1 A cluster of servers interconnected by a high-bandwidth SAN or LAN with shared I/O devices and disk arrays; the cluster acts as a single computer attached to the Internet

Single-System Image

Greg Pfister has indicated that an ideal cluster should merge multiple system images into a single-system image (SSI). Cluster designers desire a cluster operating system or some middle-ware to support SSI at various levels, including the sharing of CPUs, memory, and I/O across all cluster nodes. An SSI is an illusion created by software or hardware that presents a collection of resources as one integrated, powerful resource. SSI makes the cluster appear like a single machine to the user. A cluster with multiple system images is nothing but a collection of independent computers.

Hardware, Software, and Middleware Support

- The building blocks are computer nodes (PCs, workstations, servers, or SMP), special communication software such as PVM or MPI, and a network interface card in each computer node. Most clusters run under the Linux OS.
- The computer nodes are interconnected by a high-bandwidth network (such as Gigabit Ethernet, Myrinet, InfiniBand, etc.).
- Special cluster middleware supports are needed to create SSI or high availability (HA). Both sequential and parallel applications can run on the cluster, and special parallel environments are needed to facilitate use of the cluster resources.
- For example, distributed memory has multiple images. Users may want all distributed memory to be shared by all servers by forming distributed-ted shared memory (DSM). Many SSI features are expensive or difficult to achieve at various cluster operational levels. Instead of achieving SSI, many clusters are loosely coupled machines. Using virtualization, one can build many virtual clusters dynamically, upon user demand.

Major Cluster Design Issues

- Unfortunately, a cluster-wide OS for complete resource sharing is not available yet. Middleware or OS extensions were developed at the user space to achieve SSI at selected functional levels.
- Without this middleware, cluster nodes cannot work together effectively to achieve cooperative computing. The software environments and applications must rely on the middleware to achieve high performance.

1.1.2. Grid Computing Infrastructures

- Internet services such as the Telnet command enables a local computer to connect to a remote computer. A web service such as HTTP enables remote access of remote web pages.
- Grid computing is envisioned to allow close interaction among applications running on distant computers simultaneously.
- Forbes Magazine has projected the global growth of the IT-based economy from \$1 trillion in 2001 to \$20 trillion by 2015.
- The evolution from Internet to web and grid services is certainly playing a major role in this growth.

Features	Functional Characteristics	Feasible Implementations
Availability and Support	Hardware and Software support for sustained HA in cluster	Fail-over,Fail-back,check pointing,Rollback recovery,non-stop OS
Hardware Fault Tolerance	Automated failure management to eliminate all single points of failure.	Component redundancy,hot swapping,RAID,multiple power supplies etc
Single System Image(SSI)	Achieving SSI at functional level with hardware and software support,middleware or OS extensions.	Hardware mechanisms or middleware support to achieve DSM at coherent cache level.
Efficient Communication	To reduce message-passing system overhead and hide latencies.	Fast message passing,active messages,enhanced MPI library etc.
Cluster-Wide Job Management	Using a global job management system with better scheduling and monitoring.	Application of single-job management systems such as LSF,Codine etc.
Dynamic Load Balancing	Balancing the workload of all processing nodes along with failure recovery	Workload monitoring,process migration,job replication and gang scheduling etc.
Scalability and Programmability	Adding more servers to a cluster or adding more cluster to a grid as the workload or data set increases	Use of scalable interconnect,performance monitoring,distributed execution environment and better software tools

Table 1.2 Critical Cluster Design Issues and Feasible Implementations**Computational Grids**

- Like an electric utility power grid, a computing grid offers an infrastructure that couples computers, software/middleware, special instruments, and people and sensors together.
- The grid is often constructed across LAN, WAN, or Internet backbone networks at a regional, national, or global scale. Enterprises or organizations present grids as integrated computing resources.
- They can also be viewed as virtual platforms to support virtual organizations. The computers used in a grid are primarily workstations, servers, clusters, and supercomputers. Personal computers, laptops, and PDAs can be used as access devices to a grid system.
- Figure 1.2 shows an example computational grid built over multiple resource sites owned by different organizations.
- The resource sites offer complementary computing resources, including workstations, large servers, a mesh of processors, and Linux clusters to satisfy a chain of computational needs.
- The grid is built across various IP broadband networks including LANs and WANs already used by enterprises or organizations over the Internet. The grid is presented to users as an integrated resource pool as shown in the upper half of the figure 1.2.
- At the server end, the grid is a network. At the client end, we see wired or wireless terminal devices. The grid integrates the computing, communication, contents, and transactions as rented services. Enterprises and consumers form the user base, which then defines the usage trends and service characteristics.

Grid Families

- Grid technology demands new distributed computing models, software/middleware support, network protocols, and hardware infrastructures. National grid projects are followed by industrial grid platform development by IBM, Microsoft, Sun, HP, Dell, Cisco, EMC, Platform Computing, and others.
- New grid service providers (GSPs) and new grid applications have emerged rapidly, similar to the growth of Internet and web services in the past two decades. In Table 1.4, grid systems are classified in essentially two categories: computational or data grids and P2P grids.

Design Issues		Computational and Data Grids	P2P Grids
Grid	Application	Distributed super	Open grid with P2P flexibility, resource from

Reported	computing,National Grid Initiatives etc.	client machines.
Representative Systems	Tera Grid built in US,ChinaGrid in China and E-Science grid built in UK.	JXTA,FightAid@home,SETI@home
Development Lessons	Learned restricted user groups,middleware bugs,protocols to acquire resources.	Unreliable user-contributed resources,limited to a few apps

Table 1.4 Two Grid Computing Infrastructures and Representative Systems

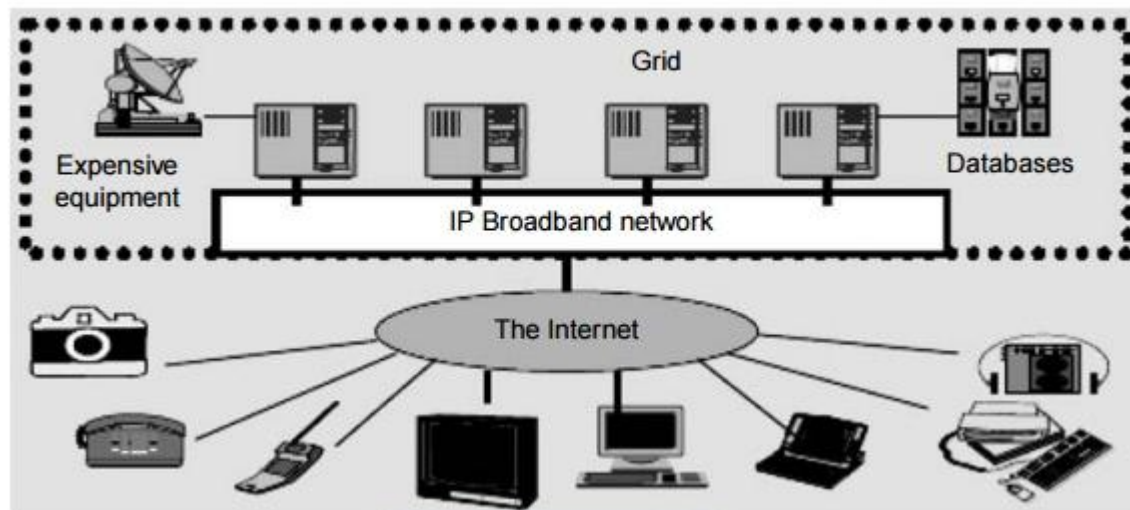


Figure 1.2 Computational grid or data grid providing computing utility, data, and information services through resource sharing and cooperation among participating organizations

1.1.3 Peer-to-Peer Network Families

- An example of a well-established distributed system is the client-server architecture. In this scenario, client machines (PCs and workstations) are connected to a central server for compute, e-mail, file access, and database applications.
- The P2P architecture offers a distributed model of networked systems. First, a P2P network is client-oriented instead of server-oriented.

P2P Systems

- In a P2P system, every node acts as both a client and a server, providing part of the system resources. Peer machines are simply client computers connected to the Internet. All client machines act autonomously to join or leave the system freely.

- This implies that no master-slave relationship exists among the peers. No central coordination or central database is needed. In other words, no peer machine has a global view of the entire P2P system.
- The system is self-organizing with distributed control. Figure 1.3 shows the architecture of a P2P network at two abstraction levels. Initially, the peers are totally unrelated.
- Each peer machine joins or leaves the P2P network voluntarily. Only the participating peers form the physical network at any time. Unlike the cluster or grid, a P2P network does not use a dedicated interconnection network. The physical network is simply an ad hoc network formed at various Internet domains randomly using the TCP/IP and NAI protocols.
- Thus, the physical network varies in size and topology dynamically due to the free membership in the P2P network.

Overlay Networks

- Data items or files are distributed in the participating peers. Based on communication or file-sharing needs, the peer IDs form an overlay network at the logical level.
- This overlay is a virtual network formed by mapping each physical machine with its ID, logically, through a virtual mapping as shown in Figure 1.3.

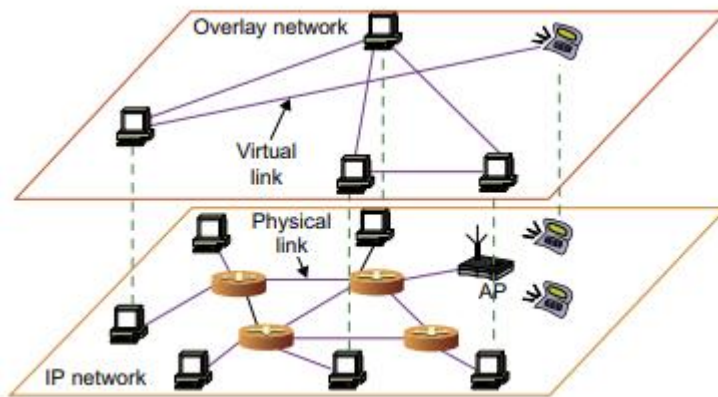


Figure 1.3 The structure of a P2P system by mapping a physical IP network to an overlay network built with virtual links

- When a new peer joins the system, its peer ID is added as a node in the overlay network. When an existing peer leaves the system, its peer ID is removed from the overlay network automatically.
- Therefore, it is the P2P overlay network that characterizes the logical connectivity among the peers.

- There are two types of overlay networks: unstructured and structured. An unstructured overlay network is characterized by a random graph. There is no fixed route to send messages or files among the nodes.
- Often, flooding is applied to send a query to all nodes in an unstructured overlay, thus resulting in heavy network traffic and nondeterministic search results.
- Structured overlay networks follow certain connectivity topology and rules for inserting and removing nodes (peer IDs) from the overlay graph.
- Routing mechanisms are developed to take advantage of the structured overlays.

P2P Application Families

- Based on application, P2P networks are classified into four groups, as shown in Table 1.3. The first family is for distributed file sharing of digital contents (music, videos, etc.) on the P2P network.
- This includes many popular P2P networks such as Gnutella, Napster, and BitTorrent, among others. Collaboration P2P networks include MSN or Skype chatting, instant messaging, and collaborative design, among others.
- The third family is for distributed P2P computing in specific applications. For example, SETI@home provides 25 Tflops of distributed computing power, collectively, over 3 million Internet host machines.
- Other P2P platforms, such as JXTA, .NET, and FightingAID@home, support naming, discovery, communication, security, and resource aggregation in some P2P applications.

System Features	Distributed File Sharing	Collaborative Platform	Distributed Computing	P2P Platform
Attractive Applications	Content distribution of MP3 music, Video, open software etc.	Instant messaging, collaborative design and gaming.	Scientific exploration and social networking.	Open networks for public resources.
Operational Problems	Loose security and serious online copyright violations	Lack of trust, distributed by spam, privacy and peer collusion.	Security holes, selfish partners and peer collusion.	Lack of standards or protection protocols.
Example Systems	Gnutella, Napster, eMule, BitTorrent, Aimster, KaZaA etc.	ICQ, AIM, Groove, Magi, Multiplayer Game, Skype etc.	SETI@home, GENOME@home etc	JXTA, .NET, FightingAid@home etc.

Table: Major Categories of P2P Network Families

P2P Computing Challenges

- P2P computing faces three types of heterogeneity problems in hardware, software, and network requirements. There are too many hardware models and architectures to select from; incompatibility exists between software and the OS; and different network connections and protocol make it too complex to apply in real applications.
- We need system scalability as the workload increases. System scaling is directly related to performance and bandwidth. P2P networks do have these properties. Data location is also important to affect collective performance.
- Data locality, network proximity, and interoperability are three design objectives in distributed P2P applications.
- P2P performance is affected by routing efficiency and self-organization by participating peers.
- Fault tolerance, failure management, and load balancing are other important issues in using overlay networks. Lack of trust among peers poses another problem. Peers are strangers to one another.
- Security, privacy, and copyright violations are major worries by those in the industry in terms of applying P2P technology in business applications .
- In a P2P network, all clients provide resources including computing power, storage space, and I/O bandwidth.
- The distributed nature of P2P networks also increases robustness, because limited peer failures do not form a single point of failure. By replicating data in multiple peers, one can easily lose data in failed nodes.

1.2.NIST CLOUD COMPUTING REFERENCE ARCHITECTURE

A model of computation and data storage based on “pay as you go” access to “unlimited” remote data center capabilities. A cloud infrastructure provides a framework to manage scalable, reliable, on-demand access to applications. Cloud services provide the “invisible” backend to many of our mobile applications. High level of elasticity in consumption.

NIST Cloud Definition:

The National Institute of Standards and Technology (NIST) defines cloud computing as a "pay-per-use model for enabling available, convenient and on-demand network access to a shared pool of

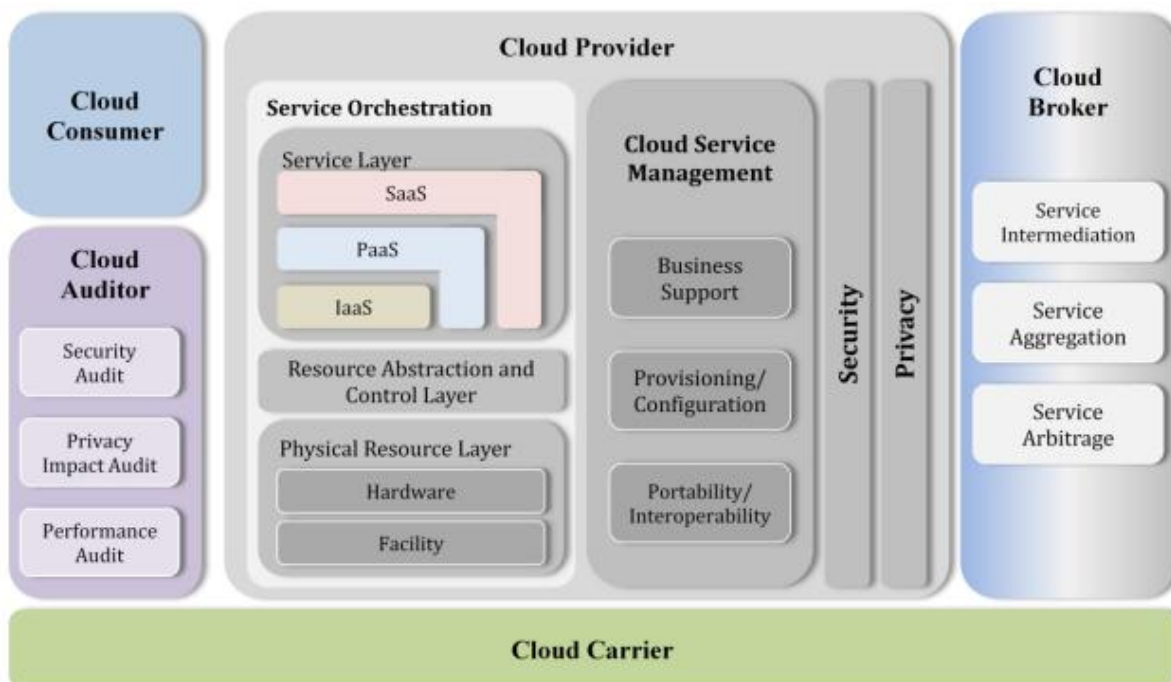
configurable computing resources (e.g., networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

1.3.Architecture

There are five major actors in NIST cloud computing reference architecture as shown in above figure.

1. Cloud Consumer.
2. Cloud Provider.
3. Cloud Carrier.
4. Cloud Auditor.
5. Cloud Broker.

Each actor is an entity may be a person or an organization that participates in a transaction or process and/or performs tasks in cloud computing. Activities and roles perform by each actor is discussed here one by one.



1. Cloud Consumer

- Cloud consumer is the main participants of cloud computing environment.
- A cloud consumer is a person or organization that use the cloud services such as SaaS, PaaS and IaaS.

- A cloud consumer browses the service catalog provided by a cloud provider, cloud consumer requests the appropriate service.
- Cloud provider sets up cloud environment for the service and make a contracts with the cloud consumer for the use of the service.
- Cloud consumers need cloud Service Level Agreement(SLA).
- SLA act as a agreement for technical performance requirements provided by a cloud provider.
- Some terms and conditions regarding the quality of service, security, remedies for performance failures are mentioned in the SLA.
- Software as a service applications in the cloud are made accessible via a network to the SaaS consumers.
- Platform as a service can also be employ by the consumer the tools to develop, test, deploy and manage the applications hosted in a cloud environment.
- PaaS consumers can be application developers who design and implement application software in software company.
- PaaS consumer may be application testers who run and test applications in cloud-based environments, application deployers who publish applications into the cloud,
- PaaS may be a application administrators who configure and monitor application performance on a platform.
- Cloud Consumers of Infrastructure as a service have access to different hardware resources like virtual computers, network devices such as router, storage media and other fundamental computing resource.
- The consumers of Infrastructure as a service may be system developers, system administrators and IT managers who creates, install, manage and monitor the services for IT infrastructure operations.

2. Cloud Provider

- A cloud provider is responsible for making a service available to the cloud consumer. Cloud provider may be a person , team or an organization.
- A Cloud Provider maintain and manages the different cloud computing services for the consumer and makes arrangement to deliver the cloud services to the Cloud Consumers suing network access or internet.
- In context to Software as a Service Cloud provider is responsible for deploys, configuring, maintaining and updating the operation of the software applications on a cloud infrastructure so that the services are provisioned as per the required levels by the cloud consumers.

- The major responsibilities of cloud provider in context to software as a service are to manage , control the applications and overall infrastructure.
- In context to Platform as a Service, the Cloud Provider manages the computing infrastructure for the platform and runs the cloud software that provides the components of the platform. These components may be software execution stack, databases and some other components that act as middleware.
- The PaaS Cloud Provider generally supports the development, deployment and management process of the Platform as a Service.
- Some integrated tools like IDE, SDK, development version of cloud software, deployment and management are also the part of Platform as a Service.
- Physical computing resources such as servers, networks, storage and hosting infrastructure are also maintain and manage by the cloud provider for the consumer of Infrastructure as a Service.

3. Cloud Auditor

- A cloud auditor is a dedicated team of technically skilled person that can perform an independent examination or review of cloud service controls with the intent to express strength and weakness of the process and some suggestion or improvement.
- Audits are performed to verify the standards of services after checking the evidence.
- Major role of a cloud auditor is to evaluate the services provided by a cloud provider against the parameters such as security controls, privacy impact and performance etc.
- To perform the audit of security a cloud auditor do the assessment of the security controls in the information system to determine the extent to which the controls are implemented accurately and operating as per expectation and producing the desired outcome with respect to the security requirements for the system.

4. Cloud Broker

- A cloud broker manages the delivery of cloud services , their performance and use.
- A cloud broker negotiates relationships between cloud providers and cloud consumers.

In general, a cloud broker involves in three types of activities which are as follow:

Service Intermediation

- A cloud broker may enhances a given service by improving some specific capability and providing value-added services to cloud consumers.
- The improvement may be related to managing the access to cloud services, identity management, performance reporting, enhanced security, etc.

Service Aggregation

- Services aggregation can be seems as combining and integrating multiple services into one or some more new services.
- The broker ensures the data movement between the cloud consumer and multiple cloud providers in secure manner.
- A cloud broker also provides the data integration.

Service Arbitrage

- Service arbitrage is very similar to service aggregation but there is a little bit difference also.
- In service arbitrage the services to be aggregated are not fixed in advance.
- In Service arbitrage a broker has the flexibility to select the services from multiple agencies.
- The cloud broker, for example, can use a credit-scoring service to measure and select an agency with the best score.

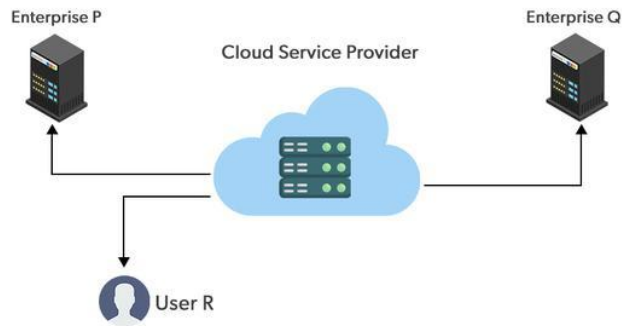
5. Cloud Carrier

- Cloud Carrier is another important actors in NIST cloud computing reference architecture.
- Role of cloud carrier is to provide the connectivity and transport of cloud services between cloud consumers and cloud providers.
- Cloud carriers provide access to consumers through network, telecommunication and other access devices.
- For example- cloud consumers can obtain cloud services through network access devices, such as computers, laptops, mobile phones, mobile Internet devices.

1.3.CLOUD DEPLOYMENT MODELS

Cloud Deployment Model functions as a virtual computing environment with a deployment architecture that varies depending on the amount of data you want to store and who has access to the infrastructure. The cloud deployment model identifies the specific type of cloud environment based on ownership, scale, and access, as well as the cloud's nature and purpose. Different types of cloud computing deployment models are described below.

- a) Public Cloud
- b) Private Cloud
- c) Hybrid Cloud
- d) Community Cloud
- e) Multi-Cloud

a)Public Cloud

- The public cloud makes it possible for anybody to access systems and services. The public cloud may be less secure as it is open to everyone.
- The public cloud is one in which cloud infrastructure services are provided over the internet to the general people or major industry groups.
- The infrastructure in this cloud model is owned by the entity that delivers the cloud services, not by the consumer.
- It is a type of cloud hosting that allows customers and users to easily access systems and services.
- In this arrangement, storage backup and retrieval services are given for free, as a subscription, or on a per-user basis. For example, Google App Engine etc.

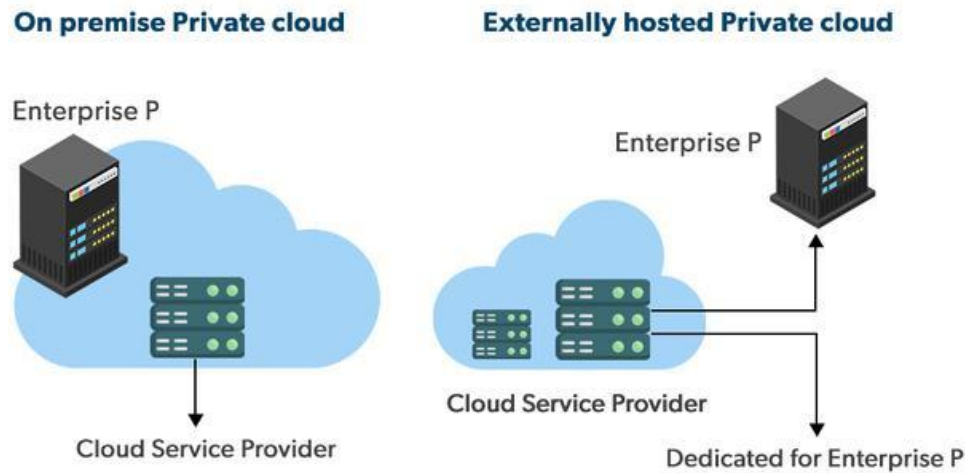
Advantages :

- Minimal Investment: Because it is a pay-per-use service, there is no substantial upfront fee, making it excellent for enterprises that require immediate access to resources.
- No setup cost: The entire infrastructure is fully subsidized by the cloud service providers, thus there is no need to set up any hardware.
- Infrastructure Management is not required: Using the public cloud does not necessitate infrastructure management.
- No maintenance: The maintenance work is done by the service provider (not users).
- Dynamic Scalability: To fulfill your company's needs, on-demand resources are accessible.

Disadvantages :

- Less secure: Public cloud is less secure as resources are public so there is no guarantee of high-level security.
- Low customization: It is accessed by many public so it can't be customized according to personal requirements.

b) Private Cloud



- The private cloud deployment model is the exact opposite of the public cloud deployment model. It's a one-on-one environment for a single user (customer).
- There is no need to share your hardware with anyone else. The distinction between private and public clouds is in how you handle all of the hardware.
- It is also called the “internal cloud” & it refers to the ability to access systems and services within a given border or organization.
- The cloud platform is implemented in a cloud-based secure environment that is protected by powerful firewalls and under the supervision of an organization's IT department. The private cloud gives greater flexibility of control over cloud resources.

Advantages:

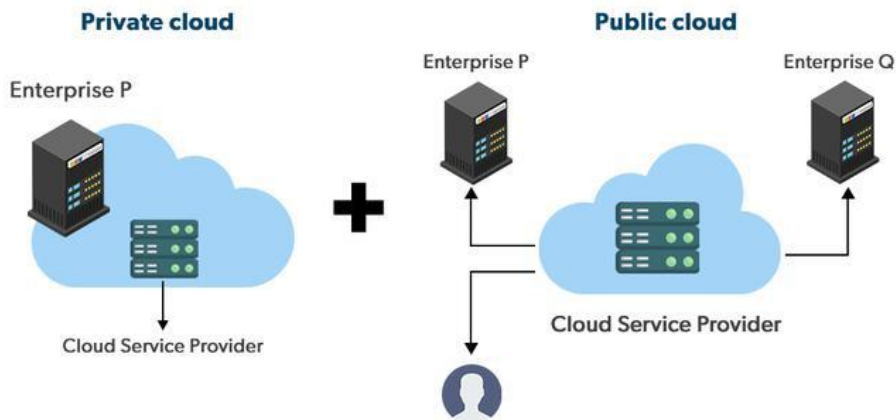
- Better Control: You are the sole owner of the property. You gain complete command over service integration, IT operations, policies, and user behavior.
- Data Security and Privacy: It's suitable for storing corporate information to which only authorized staff have access. By segmenting resources within the same infrastructure, improved access and security can be achieved.
- Supports Legacy Systems: This approach is designed to work with legacy systems that are unable to access the public cloud.
- Customization: Unlike a public cloud deployment, a private cloud allows a company to tailor its solution to meet its specific needs.

Disadvantages:

- Less scalable: Private clouds are scaled within a certain range as there is less number of clients.

- Costly: Private clouds are more costly as they provide personalized facilities.

c) Hybrid Cloud



- By bridging the public and private worlds with a layer of proprietary software, hybrid cloud computing gives the best of both worlds. With a hybrid solution, you may host the app in a safe environment while taking advantage of the public cloud's cost savings.
- Organizations can move data and applications between different clouds using a combination of two or more cloud deployment methods, depending on their needs.

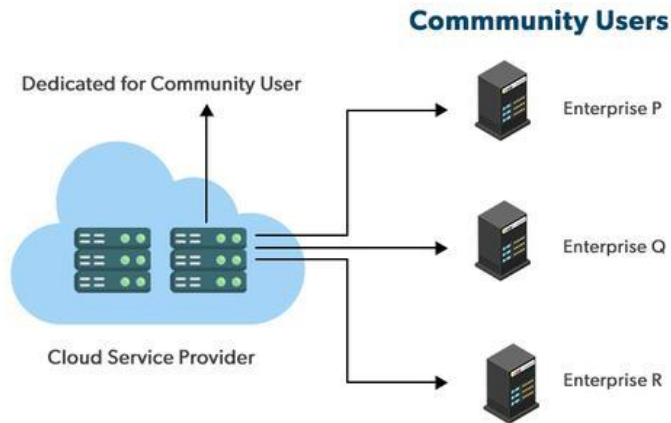
Advantages:

- Flexibility and control: Businesses with more flexibility can design personalized solutions that meet their particular needs.
- Cost: Because public clouds provide scalability, you'll only be responsible for paying for the extra capacity if you require it.
- Security: Because data is properly separated, the chances of data theft by attackers are considerably reduced.

Disadvantages

- Difficult to manage: Hybrid clouds are difficult to manage as it is a combination of both public and private cloud. So, it is complex.
- Slow data transmission: Data transmission in the hybrid cloud takes place through the public cloud so latency occurs.

d) Community Cloud



- It allows systems and services to be accessible by a group of organizations. It is a distributed system that is created by integrating the services of different clouds to address the specific needs of a community, industry, or business.
- The infrastructure of the community could be shared between the organization which has shared concerns or tasks. It is generally managed by a third party or by the combination of one or more organizations in the community.

Advantages

- **Cost Effective:** It is cost-effective because the cloud is shared by multiple organizations or communities.
- **Security:** Community cloud provides better security.
- **Shared resources:** It allows you to share resources, infrastructure, etc. with multiple organizations.
- **Collaboration and data sharing:** It is suitable for both collaboration and data sharing.

Disadvantages

- **Limited Scalability:** Community cloud is relatively less scalable as many organizations share the same resources according to their collaborative interests.
- **Rigid in customization:** As the data and resources are shared among different organizations according to their mutual interests if an organization wants some changes according to their needs they cannot do so because it will have an impact on other organizations.

A Comparative Analysis of Cloud Deployment Models

Important Factors to Consider	Public	Private	Community	Hybrid
Setup and ease of use	Easy	Requires professional IT Team	Requires professional IT Team	Requires professional IT Team
Data Security and Privacy	Low	High	Very High	High
Scalability and flexibility	High	High	Fixed requirements	High
Cost-Effectiveness	Most affordable	Most expensive	Cost is distributed among members	Cheaper than private but more expensive than public
Reliability	Low	High	Higher	High

1.4.CLOUD SERVICE MODELS

There are the following three types of cloud service models –

- i. Infrastructure as a Service (IaaS)
- ii. Platform as a Service (PaaS)
- iii. Software as a Service (SaaS)

i)Infrastructure as a Service (IaaS)

- Infrastructure as a service (IaaS) is a service model that delivers computer infrastructure on an outsourced basis to support various operations.
- Typically IaaS is a service where infrastructure is provided as outsourcing to enterprises such as networking equipment, devices, database, and web servers.
- It is also known as Hardware as a Service (HaaS). IaaS customers pay on a per-user basis, typically by the hour, week, or month. Some providers also charge customers based on the amount of virtual machine space they use.
- It simply provides the underlying operating systems, security, networking, and servers for developing such applications, and services, and deploying development tools, databases, etc.

Advantages of IaaS:

- **Cost-Effective:** Eliminates capital expense and reduces ongoing cost and IaaS customers pay on a per-user basis, typically by the hour, week, or month.
- **Website hosting:** Running websites using IaaS can be less expensive than traditional web hosting.
- **Security:** The IaaS Cloud Provider may provide better security than your existing software.
- **Maintenance:** There is no need to manage the underlying data center or the introduction of new releases of the development or underlying software. This is all handled by the IaaS Cloud Provider.

Disadvantages of IaaS :

- **Limited control over infrastructure:** IaaS providers typically manage the underlying infrastructure and take care of maintenance and updates, but this can also mean that users have less control over the environment and may not be able to make certain customizations.
- **Security concerns:** Users are responsible for securing their own data and applications, which can be a significant undertaking.
- **Limited access:** Cloud computing may not be accessible in certain regions and countries due to legal policies.

Example: DigitalOcean, Linode, Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine (GCE), Rackspace, and Cisco Metacloud, VMware..

ii) Platform as a Service

- PaaS is a category of cloud computing that provides a platform and environment to allow developers to build applications and services over the internet.
- PaaS services are hosted in the cloud and accessed by users simply via their web browser.
- A PaaS provider hosts the hardware and software on its own infrastructure. As a result, PaaS frees users from having to install in-house hardware and software to develop or run a new application.
- Thus, the development and deployment of the application take place independent of the hardware.
- The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly configuration settings for the application-hosting environment.

Advantages of PaaS:

- Simple and convenient for users: It provides much of the infrastructure and other IT services, which users can access anywhere via a web browser.
- Cost-Effective: It charges for the services provided on a per-use basis thus eliminating the expenses one may have for on-premises hardware and software.
- Efficiently managing the lifecycle: It is designed to support the complete web application lifecycle: building, testing, deploying, managing, and updating.
- Efficiency: It allows for higher-level programming with reduced complexity thus, the overall development of the application can be more effective.

The various companies providing Platform as a service are Amazon Web services Elastic Beanstalk, Salesforce, Windows Azure, Google App Engine, cloud Bees and IBM smart cloud.

Disadvantages of Paas:

- Limited control over infrastructure: PaaS providers typically manage the underlying infrastructure and take care of maintenance and updates, but this can also mean that users have less control over the environment and may not be able to make certain customizations.
- Dependence on the provider: Users are dependent on the PaaS provider for the availability, scalability, and reliability of the platform, which can be a risk if the provider experiences outages or other issues.
- Limited flexibility: PaaS solutions may not be able to accommodate certain types of workloads or applications, which can limit the value of the solution for certain organizations.

iii)Software as a Service(SaaS)

- Software-as-a-Service (SaaS) is a way of delivering services and applications over the Internet. Instead of installing and maintaining software, we simply access it via the Internet, freeing ourselves from the complex software and hardware management.
- It removes the need to install and run applications on our own computers or in the data centers eliminating the expenses of hardware as well as software maintenance.
- SaaS provides a complete software solution that you purchase on a pay-as-you-go basis from a cloud service provider.

Advantages

- Cost-Effective: Pay only for what you use.
- Reduced time: Users can run most SaaS apps directly from their web browser without needing to download and install any software. This reduces the time spent in installation and configuration and can reduce the issues that can get in the way of the software deployment.

- Accessibility: We can Access app data from anywhere.
- Automatic updates: Rather than purchasing new software, customers rely on a SaaS provider to automatically perform the updates.
- Scalability: It allows the users to access the services and features on-demand.

The various companies providing Software as a service are Cloud9 Analytics, Salesforce.com, Cloud Switch, Microsoft Office 365, Big Commerce, Eloqua, dropBox, and Cloud Tran.

Disadvantages of Saas :

- Limited customization: SaaS solutions are typically not as customizable as on-premises software
- Dependence on internet connectivity: SaaS solutions are typically cloud-based, which means that they require a stable internet connection to function properly.
- Security concerns: SaaS providers are responsible for maintaining the security of the data stored on their servers, but there is still a risk of data breaches or other security incidents.
- Limited control over data: SaaS providers may have access to a user's data, which can be a concern for organizations that need to maintain strict control over their data for regulatory or Example: BigCommerce, Google Apps, Salesforce, Dropbox, ZenDesk, Cisco WebEx, ZenDesk, Slack, and GoToMeeting.

1.5. CLOUD INFRASTRUCTURE: ARCHITECTURAL DESIGN OF COMPUTE AND STORAGE CLOUDS

1.5.1 A Generic Cloud Architecture Design

- An Internet cloud is envisioned as a public cluster of servers provisioned on demand to perform collective web services or distributed applications using data-center resources.

- ❖ Cloud Platform Design Goals
- ❖ Enabling Technologies for Clouds
- ❖ A Generic Cloud Architecture

Cloud Platform Design Goals

- ✓ Scalability
- ✓ Virtualization
- ✓ Efficiency
- ✓ Reliability

✓ Security

Cloud management receives the user request and finds the correct resources. Cloud calls the provisioning services which invoke the resources in the cloud. Cloud management software needs to support both physical and virtual machines

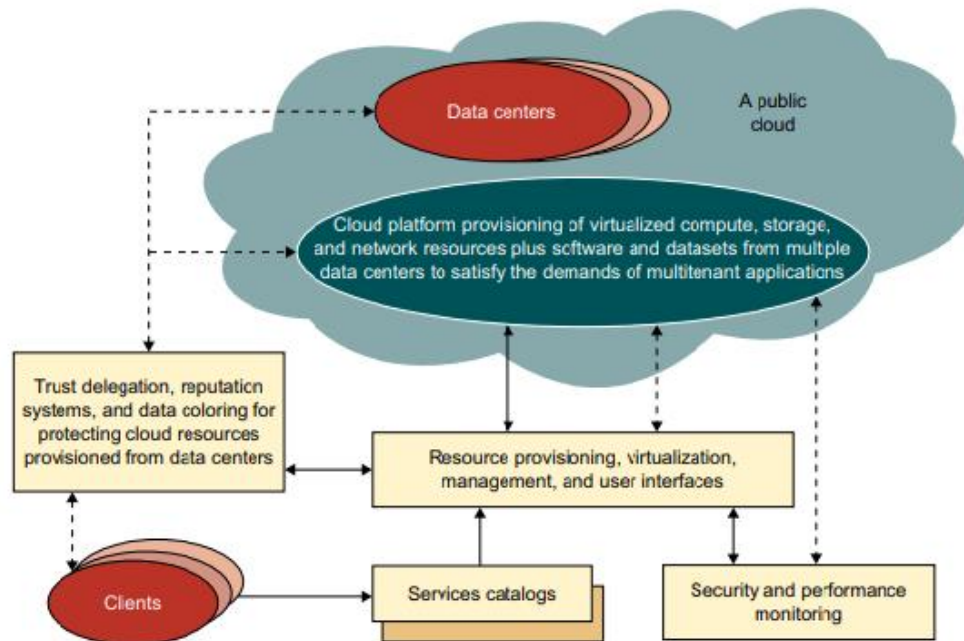
Enabling Technologies for Clouds

- Cloud users are able to demand more capacity at peak demand, reduce costs, experiment with new services, and remove unneeded capacity.
- Service providers can increase system utilization via multiplexing, virtualization and dynamic resource provisioning.
- Clouds are enabled by the progress in hardware, software and networking technologies
- Cloud users are able to demand more capacity at peak demand, reduce costs, experiment with new services, and remove unneeded capacity.
- Service providers can increase system utilization via multiplexing, virtualization and dynamic resource provisioning.
- Clouds are enabled by the progress in hardware, software and networking

Technologies

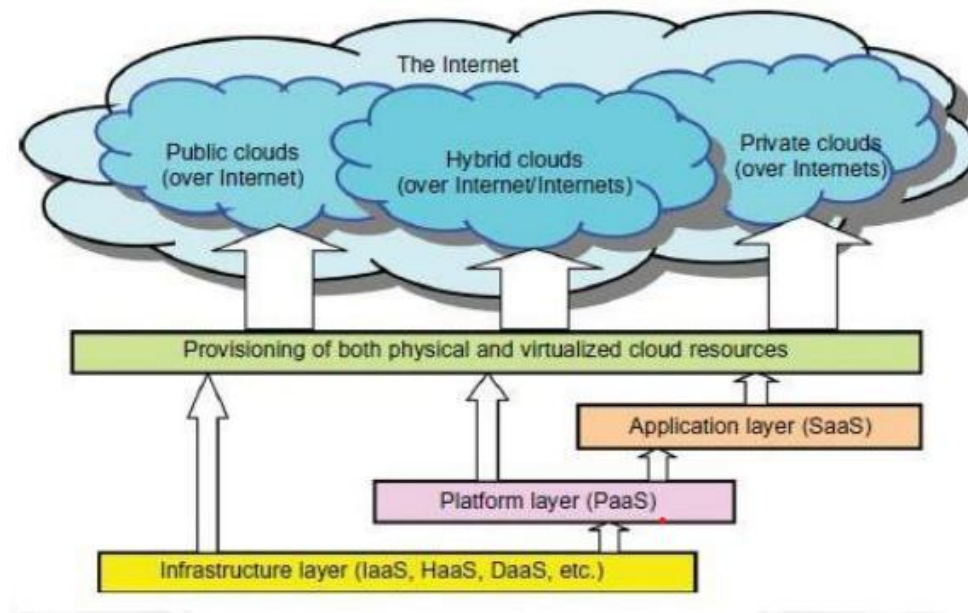
Table Cloud-Enabling Technologies in Hardware, Software, and Networking	
Technology	Requirements and Benefits
Fast platform deployment	Fast, efficient, and flexible deployment of cloud resources to provide dynamic computing environment to users
Virtual clusters on demand	Virtualized cluster of VMs provisioned to satisfy user demand and virtual cluster reconfigured as workload changes
Multitenant techniques	SaaS for distributing software to a large number of users for their simultaneous use and resource sharing if so desired
Massive data processing	Internet search and web services which often require massive data processing, especially to support personalized services
Web-scale communication	Support for e-commerce, distance education, telemedicine, social networking, digital government, and digital entertainment applications
Distributed storage	Large-scale storage of personal records and public archive information which demands distributed storage over the clouds
Licensing and billing services	License management and billing services which greatly benefit all types of cloud services in utility computing

A Generic Cloud Architecture



- The Internet cloud is envisioned as a massive cluster of servers.
- Servers are provisioned on demand to perform collective web services using data-center resources. The cloud platform is formed dynamically by provisioning or deprovisioning servers, software, and database resources.
- Servers in the cloud can be physical machines or VMs.
- User interfaces are applied to request services.
- The cloud computing resources are built into the data centers. Data centers are typically owned and operated by a third-party provider. Consumers do not need to know the underlying technologies.
- In a cloud, software becomes a service. Cloud demands a high degree of trust of massive amounts of data retrieved from large data centers.
- The software infrastructure of a cloud platform must handle all resource management and maintenance automatically.
- Software must detect the status of each node server joining and leaving.
- Cloud computing providers such as Google and Microsoft, have built a large number of data centers.
- Each data center may have thousands of servers.
- The location of the data center is chosen to reduce power and cooling costs.

1.5.2.Layered Architecture:



The architecture of a cloud is developed at three layers

- A. Infrastructure
 - B. Platform
 - C. Application
- Implemented with virtualization and standardization of hardware and software resources provisioned in the cloud.
 - The services to public, private and hybrid clouds are conveyed to users through networking support

A.Infrastructure Layer

- Foundation for building the platform layer.
- Built with virtualized compute, storage, and network resources.
- Provide the flexibility demanded by users.
- Virtualization realizes automated provisioning of resources and optimizes the infrastructure management process.

B. Platform Layer

- Foundation for implementing the application layer for SaaS applications.
- Used for general-purpose and repeated usage of the collection of software resources.

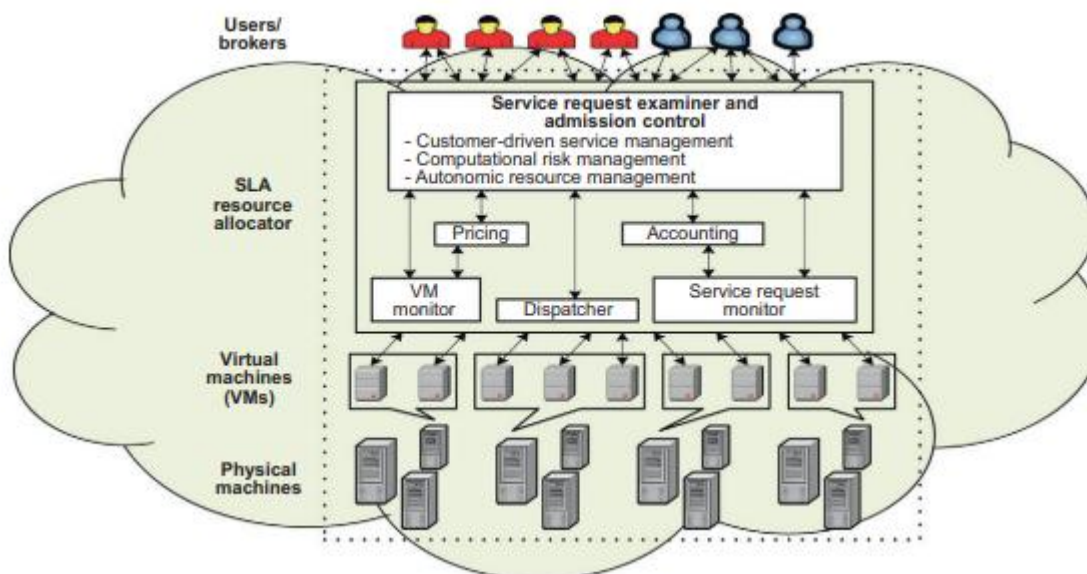
- Provides users with an environment to develop their applications, to test operation flows, and to monitor execution results and performance.
- The platform should be able to assure users that they have scalability, dependability, and security protection

C.Application Layer

- Collection of all needed software modules for SaaS applications.
- Service applications in this layer include daily office management work, such as information retrieval, document processing, and authentication services.
- The application layer is also heavily used by enterprises in business marketing and sales, consumer relationship management (CRM) and financial transactions.
- Not all cloud services are restricted to a single layer.
- Many applications may apply resources at mixed layers.
- Three layers are built from the bottom up with a dependence relationship.

Market-Oriented Cloud Architecture

- High-level architecture for supporting market-oriented resource allocation in a cloud computing environment.
- Users or brokers acting on user's behalf submit service requests to the data center.
- When a service request is first submitted, the service request examiner interprets the submitted request for QoS requirements.



- **VM Monitor:** Latest status information regarding resource availability.
- **Service Request Monitor:** Latest status information workload processing

- Pricing mechanism:Decides how service requests are charged.
- Accounting mechanism:Maintains the actual usage of resources by requests to compute the final cost.
- VM Monitor mechanism keeps track of the availability of VMs and their resource entitlements.
- Dispatcher starts the execution of accepted service requests on allocated VMs.
- Service Request Monitor mechanism keeps track of the execution progress of service requests.
- Multiple VMs can be started and stopped on demand

Quality of Service Factors

QoS parameters

- ✓ Time
- ✓ Cost
- ✓ Reliability
- ✓ Trust/security

QoS requirements cannot be static and may change over time.

1.5.3.Virtualization Support and Disaster Recover:

- One very distinguishing feature of cloud computing infrastructure is the use of system virtualization and the modification to provisioning tools.
- Virtualizations of servers on a shared cluster can consolidate web services. As the VMs are the containers of cloud services, the provisioning tools will first find the corresponding physical machines and deploy the VMs to those nodes before scheduling the service to run on the virtual nodes.
- In addition, in cloud computing, virtualization also means the resources and fundamental infrastructure are virtualized. The user will not care about the computing resources that are used for providing the services.
- Cloud users do not need to know and have no way to discover physical resources that are involved while processing a service request.
- Also, application developers do not care about some infrastructure issues such as scalability and fault tolerance (i.e., they are virtualized). Application developers focus on service logic.
- Fig1.5. shows the infrastructure needed to virtualize the servers in a data center for implementing specific cloud applications.

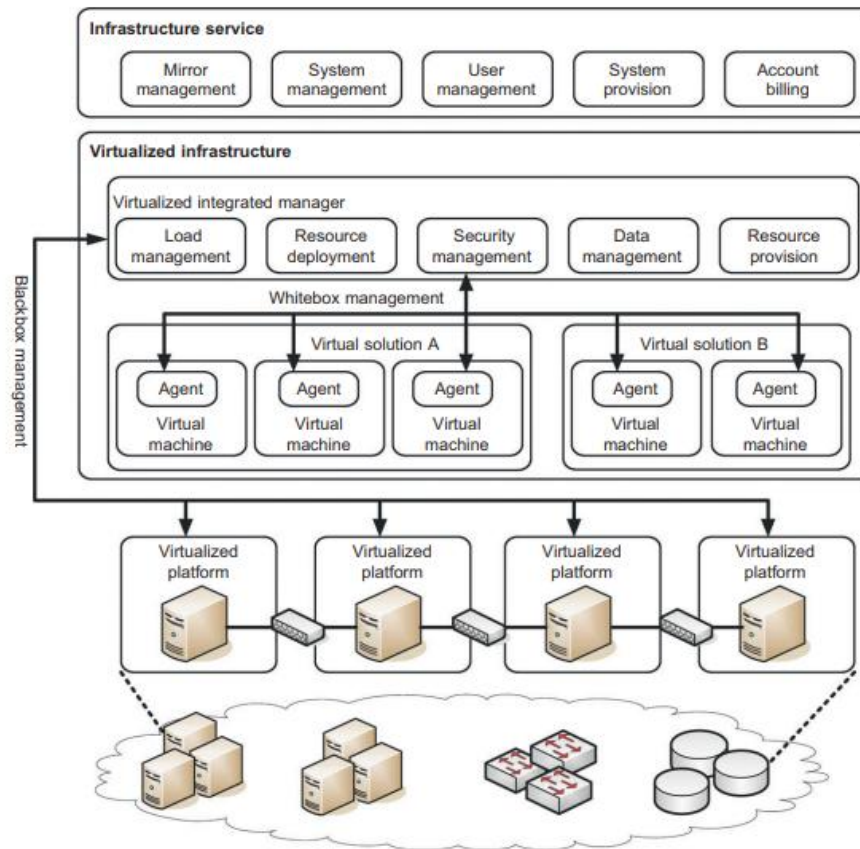


Figure.1.5 Virtualized servers, storage, and network for cloud platform construction.

1.5.3.1 Hardware Virtualization

- Cloud computing systems use virtualization software as the running environment for legacy software such as old operating systems and unusual applications.
- Virtualization software is also used as the platform for developing new cloud applications that enable developers to use any operating systems and programming environments they like.
- The development environment and deployment environment can now be the same, which eliminates some runtime problems. S
- Some cloud computing providers have used .virtualization technology to provide this service for developers.
- Currently, the VMs installed on a cloud computing platform are mainly used for hosting third-party programs. VMs provide flexible runtime services to free users from worrying about the system environment

1.5.3.2. Virtualization Support in Public Clouds

- AWS provides extreme flexibility (VMs) for users to execute their own applications.

- GAE provides limited application-level virtualization for users to build applications only based on the services that are created by Google.
- Microsoft provides programming-level virtualization (.NET virtualization) for users to build their applications. The VMware tools apply to workstations, servers, and virtual infrastructure. The Microsoft tools are used on PCs and some special servers.
- The Xen Enterprise tool applies only to Xen-based servers. Everyone is interested in the cloud; the entire IT industry is moving toward the vision of the cloud.
- Virtualization leads to HA, disaster recovery, dynamic load leveling, and rich provisioning support.

1.5.3.3 Storage Virtualization for Green Data Centers

- IT power consumption in the United States has more than doubled to 3 percent of the total energy consumed in the country.
- The large number of data centers in the country has contributed to this energy crisis to a great extent.
- More than half of the companies in the Fortune 500 are actively implementing new corporate energy policies.
- Recent surveys from both IDC and Gartner confirm the fact that virtualization had a great impact on cost reduction from reduced power consumption in physical computing systems.
- This alarming situation has made the IT industry become more energy aware. With little evolution of alternate energy resources, there is an imminent need to conserve power in all computers.
- Virtualization and server consolidation have already proven handy in this aspect.
- Green data centers and benefits of storage virtualization are considered to further strengthen the synergy of green computing.

1.5.3.4 Virtualization for IaaS

VM technology has increased in ubiquity. This has enabled users to create customized environments atop physical infrastructure for cloud computing. Use of VMs in clouds has the following distinct benefits:

- (1) System administrators consolidate workloads of underutilized servers in fewer servers;
- (2) VMs have the ability to run legacy code without interfering with other APIs;
- (3) VMs can be used to improve security through creation of sandboxes for running applications with questionable reliability;

- (4) Virtualized cloud platforms can apply performance isolation, letting providers offer some guarantees and better QoS to customer applications

1.5.3.5 VM Cloning for Disaster Recovery

- VM technology requires an advanced disaster recovery scheme. One scheme is to recover one physical machine by another physical machine.
- The second scheme is to recover one VM by another VM. Traditional disaster recovery from one physical machine to another is rather slow, complex, and expensive.
- Total recovery time is attributed to the hardware configuration, installing and configuring the OS, installing the backup agents, and the long time to restart the physical machine.
- To recover a VM platform, the installation and configuration times for the OS and backup agents are eliminated.
- Virtualization aids in fast disaster recovery by VM encapsulation.
- The cloning of VMs offers an effective solution. The idea is to make a clone VM on a remote server for every running VM on a local server.
- Among all the clone VMs, only one needs to be active. The remote VM should be in a suspended mode.
- A cloud control center should be able to activate this clone VM in case of failure of the original VM, taking a snapshot of the VM to enable live migration in a minimal amount of time.
- The migrated VM can run on a shared Internet connection. Only updated data and modified states are sent to the suspended VM to update its state. The
- Recovery Property Objective (RPO) and Recovery Time Objective (RTO) are affected by the number of snapshots taken. Security of the VMs should be enforced during live migration of VMs.

1.6. ARCHITECTURAL DESIGN CHALLENGES

Challenge 1: Service Availability and Data Lock-in Problem

- The management of a cloud service by a single company is often the source of single points of failure.
- To achieve HA, one can consider using multiple cloud providers.
- Even if a company has multiple data centers located in different geographic regions, it may have common software infrastructure and accounting systems.
- Therefore, using multiple cloud providers may provide more protection from failures.
- Another availability obstacle is distributed denial of service (DDoS) attacks.

- Criminals threaten to cut off the incomes of SaaS providers by making their services unavailable.
- Some utility computing services offer SaaS providers the opportunity to defend against DDoS attacks by using quick scale ups.
- Software stacks have improved interoperability among different cloud platforms, but the APIs itself are still proprietary. Thus, customers cannot easily extract their data and programs from one site to run on another.
- The obvious solution is to standardize the APIs so that a SaaS developer can deploy services and data across multiple cloud providers.
- This will rescue the loss of all data due to the failure of a single company.
- In addition to mitigating data lock-in concerns, standardization of APIs enables a new usage model in which the same software infrastructure can be used in both public and private clouds.
- Such an option could enable surge computing, in which the public cloud is used to capture the extra tasks that cannot be easily run in the data center of a private cloud.

Challenge 2: Data Privacy and Security Concerns

- Current cloud offerings are essentially public (rather than private) networks, exposing the system to more attacks.
- Many obstacles can be overcome immediately with well understood technologies such as encrypted storage, virtual LANs, and network middle boxes (e.g., firewalls, packet filters).
- For example, the end user could encrypt data before placing it in a cloud.
- Many nations have laws requiring SaaS providers to keep customer data and copyrighted material within national boundaries.
- Traditional network attacks include buffer overflows, DoS attacks, spyware, malware, rootkits, Trojan horses, and worms.
- In a cloud environment, newer attacks may result from hypervisor malware, guest hopping and hijacking or VM rootkits.
- Another type of attack is the man-in-the-middle attack for VM migrations.
- In general, passive attacks steal sensitive data or passwords.
- On the other hand, Active attacks may manipulate kernel data structures which will cause major damage to cloud servers.

Challenge 3: Unpredictable Performance and Bottlenecks

- Multiple VMs can share CPUs and main memory in cloud computing, but I/O sharing is problematic.
- For example, to run 75 EC2 instances with the STREAM benchmark requires a mean bandwidth of 1,355 MB/second.
- However, for each of the 75 EC2 instances to write 1 GB files to the local disk requires a mean disk write bandwidth of only 55 MB/second.
- This demonstrates the problem of I/O interference between VMs.
- One solution is to improve I/O architectures and operating systems to efficiently virtualize interrupts and I/O channels. Internet applications continue to become more data intensive.
- If we assume applications to be pulled apart across the boundaries of clouds, this may complicate data placement and transport.
- Cloud users and providers have to think about the implications of placement and traffic at every level of the system, if they want to minimize costs.
- This kind of reasoning can be seen in Amazon's development of its new CloudFront service.
- Therefore, data transfer bottlenecks must be removed, bottleneck links must be widened and weak servers should be removed.

Challenge 4: Distributed Storage and Widespread Software Bugs

- The database is always growing in cloud applications.
- The opportunity is to create a storage system that will not only meet this growth but also combine it with the cloud advantage of scaling arbitrarily up and down on demand.
- This demands the design of efficient distributed SANs.
- Data centers must meet programmer's expectations in terms of scalability, data durability and HA. Data consistence checking in SAN connected data centers is a major challenge in cloud computing.
- Large scale distributed bugs cannot be reproduced, so the debugging must occur at a scale in the production data centers.
- No data center will provide such a convenience. One solution may be a reliance on using VMs in cloud computing.
- The level of virtualization may make it possible to capture valuable information in ways that are impossible without using VMs.
- Debugging over simulators is another approach to attacking the problem, if the simulator is well designed.

Challenge 5: Cloud Scalability, Interoperability, and Standardization

- The pay as you go model applies to storage and network bandwidth; both are counted in terms of the number of bytes used.
- Computation is different depending on virtualization level.
- GAE automatically scales in response to load increases or decreases and the users are charged by the cycles used.
- AWS charges by the hour for the number of VM instances used, even if the machine is idle.
- The opportunity here is to scale quickly up and down in response to load variation, in order to save money, but without violating SLAs.
- Open Virtualization Format (OVF) describes an open, secure, portable, efficient and extensible format for the packaging and distribution of VMs.
- It also defines a format for distributing software to be deployed in VMs.
- This VM format does not rely on the use of a specific host platform, virtualization platform or guest operating system.
- The approach is to address virtual platform is agnostic packaging with certification and integrity of packaged software.
- The package supports virtual appliances to span more than one VM.
- OVF also defines a transport mechanism for VM templates and the format can apply to different virtualization platforms with different levels of virtualization.
- In terms of cloud standardization, the ability for virtual appliances to run on any virtual platform.
- The user is also need to enable VMs to run on heterogeneous hardware platform hypervisors.
- This requires hypervisor-agnostic VMs.
- And also the user need to realize cross platform live migration between x86 Intel and

Challenge 6: Software Licensing and Reputation Sharing

- Many cloud computing providers originally relied on open source software because the licensing model for commercial software is not ideal for utility computing.
- The primary opportunity is either for open source to remain popular or simply for commercial software companies to change their licensing structure to better fit cloud computing.
- One can consider using both pay for use and bulk use licensing schemes to widen the business coverage.

TWO MARK QUESTIONS AND ANSWERS**1. Define Cloud computing.**

Cloud computing is a model for delivering IT services in which resources are retrieved from the internet through web based tools and applications rather than a direct connection to the server.

2. Define Cloud.

Cloud refers to software, platform, and Infrastructure that are sold as a service. The services accessed remotely through the Internet. The cloud users can simply log on to the network without installing anything. They do not pay for hardware and maintenance. But the service providers pay for physical equipment and maintenance.

3. List the entities involved in the cloud platform.

- Users or brokers and Request examiner
- Pricing mechanism and VM Monitor mechanism
- Accounting mechanism
- Service Request Examiner and Admission Control mechanism
- Dispatcher mechanism
- Service Request Monitor mechanism

4. List the objectives of NIST Cloud Computing reference architecture

- Illustrate and understand the various level of services
- To provide technical reference
- Categorize and compare services of cloud computing
- Analysis of security, interoperability and portability

5. Mention the major actors involved in NIST reference model.

- Cloud consumer
- Cloud provider
- Cloud auditor
- Cloud broker
- Cloud carrier

6. Define service orchestration.

Service orchestration describes the automated arrangement, coordination and management of complex computing system.

7. Differentiate between Public cloud and Private Cloud.

- A public cloud is one in which the cloud infrastructure and computing resources are made available to the general public over a public network.
- A public cloud is owned by an organization selling cloud services, and serves a diverse pool of clients.
- A private cloud gives a single Cloud Consumer's organization the exclusive access to and usage of the infrastructure and computational resources.
- It may be managed either by the Cloud Consumer organization or by a third party, and may be hosted on the organization's premises (i.e. on-site private clouds) or outsourced to a hosting company (i.e. outsourced private clouds).

8. Tabulate the merits and demerits of Choosing a Community Cloud.

Merits	Demerits
Ability to easily share and collaborate	Not the right choice for every organization
Lower cost	Slow adoption to date

9. What is IaaS or HaaS?

Infrastructure or Hardware-as-a-Service (IaaS/HaaS) solutions are the most popular and developed market segment of cloud computing. They deliver customizable infrastructure on demand.

10. What is PaaS?

Platform-as-a-Service (PaaS) solutions provide a development and deployment platform for running applications in the cloud. They constitute the middleware on top of which applications are built.

11. Classify Pass Implementation

PaaS implementation classified into three wide categories:

- PaaS-I,
- PaaS-II
- PaaS-III.
-

12. What is SaaS?

Software-as-a-Service (SaaS) is a software delivery model that provides access to applications through the Internet as a Web-based service. It provides a means to free users from complex hardware and software management by offloading such tasks to third parties, which build applications accessible to multiple users through a Web browser.

13. What is SaaS 2.0?

In particular, SaaS 2.0 is focused on providing a more robust infrastructure and application platforms driven by SLAs. SaaS 2.0 will focus on the rapid achievement of business objectives.

14. List the six architectural design challenges in cloud.

- Service Availability and Data Lock-in Problem
- Data Privacy and Security Concerns
- Unpredictable Performance and Bottlenecks
- Distributed Storage and Widespread Software Bugs
- Cloud Scalability, Interoperability, and Standardization
- Software Licensing and Reputation Sharing

15. What is cloud storage?

Cloud storage means storing the data with a cloud service provider rather than on a local system. The end user can access the data stored on the cloud using an Internet link. Cloud storage has a number of advantages over traditional data storage. If the users stored some data on a cloud, they can get at it from any location that has Internet access.

16. What is Storage-as-a-Service?

The term Storage as a Service means that a third-party provider rents space on their

storage to end users who lack the budget or capital budget to pay for it on their own. It is also ideal when technical personnel are not available or have inadequate knowledge to implement and maintain that storage infrastructure.

17. Why do we need hybrid cloud

Maintain security and high performance

Run workloads where they perform best

Reduce IT cost and improve network efficiency

18. How to secure data in cloud storage?
Most systems use a combination of following techniques:

- Encryption
- Authentication processes
- Authorization practices

19. List the advantages of cloud storage.

- Storing data on the cloud is advantageous, because it allows you to protect your data in case there's a disaster.
- Having your data stored off-site can be the difference between closing your door for good or being down for a few days or weeks.
- Which storage vendor to go with can be a complex issue, and how the end user technology interacts with the cloud can be complex.

20. What is S3?

The best-known cloud storage service is Amazon's Simple Storage Service (S3), which launched in 2006. Amazon S3 is designed to make web-scale computing easier for developers. Amazon S3 provides a simple web services interface that can be used to store and retrieve any amount of data, at any time, from anywhere on the Web. It gives any developer access to the same highly scalable data storage infrastructure that Amazon uses to run its own global network of web sites.

21. List some of the advantages of cloud computing?

Unlimited Storage	Cost Efficient
Backup and Recovery	Easy Access to Information
Quick Deployment	Lower IT Infrastructure Costs
Fewer Maintenance Issues	Lower Software Costs
Instant Software Updates	increased Computing Power

Easier Group Collaboration Universal Access to Documents

22. List some of the disadvantages of cloud computing?

Technical Issues

- Less Secured
- Prone to Attack
- Limited Control

Requires Constant Internet Connection

Can Be Slow

Stored Data Might Not Be Secure

23. Mention the characteristics features of the cloud (Apr/May 2017)

On-Demand Usage

Ubiquitous Access

Multi-tenancy (Resourcing Pooling)

Elasticity (and Scalability)

Measured Usage

Resiliency

PART-B&C

1. What are the services provided by cloud with deployment model? Explain in detail (Nov/Dec 2017)
2. List the cloud deployment models and give a detailed about them (Nov/Dec 2016)
3. Explain in detail about cloud delivery model.
4. Discuss the operational and economic benefits of SaaS.
5. Give the importance of cloud computing and elaborate the different types of services offered by it. (Nov/Dec 2016)
6. Specify a scenario where PaaS can be applied
7. Identify upfront Investment in New Initiatives and what will be its benefits in cloud